Bartosz Włodarczyk

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Education

2024 – 2027 **Delft University of Technology** - Honours Student, Bachelor of Computer Science and Engineering, with a GPA of 8.9/10

2020 – 2024 **Batory High School in Warsaw** - International Baccalaureate Diploma Programme

Skills

Languages **English** - fluent, **German** - B2, **Dutch** - A2, **Polish** - native

Programming Java, SQL, Python, LaTeX, some experience in: C++, Rust

Web Dev HTML, CSS, JavaScript, TypeScript, Tailwind, Svelte(Kit), React, Next.js

Employment History

2025 – present **Teaching Assistant**, **Developer** at PRIME, TU Delft - I am creating educational resources for Mathematics courses offered in engineering programs at the university.

- [2026 – present] **Python Developer | Practica Numerical Analysis** - I am developing Python notebooks for Numerical Analysis courses, teaching students about how to apply their theoretical knowledge in practical ways. I am responsible for developing the test suite to check students' answers, and also providing student's perspective on the assignments.
- [2025 – present] **Svelte Web Developer | Open LA Applets** - I am developing web-based applets (interactive animations) used in Linear Algebra and Calculus Mathematics textbooks, and the framework used to create them. The framework is written in Svelte 5 with Tailwind CSS and TypeScript. Source code: `https://github.com/PRIME-TU-Delft/Open-LA-Applets`.

2024 – 2025 **Mathematics Tutor** - I have provided professional Mathematics tutoring to high school students, in both English and Polish. I also worked with tutoring agencies such as *Acady*.

Awards and Achievements

2024 **Best mathematician graduate award**, *Łukasiewicz award*, awarded by my high school annually to the best graduate in the field of Mathematics.

2023 **Honorable mention**, 9th annual International Mathematical Modelling Challenge, a contest organised, among others, by Freudenthal Institute.

Special recognition award, *Matużytkownicy* national mathematical modelling contest organised by Warsaw University of Technology.

2022 **Merit Award**, Singapore International Mathematical and Computational Senior Challenge organised by National University of Singapore High School - won the silver medal.

Top 10% rank in *Náboj* 2022 Juniors international mathematical competition organised, among others, by ETH Zürich and University of Cambridge.

Experience

Professional experience

2025 — **BurtCore** - is a hackathon project I worked on with a team of 5. We developed a machine learning model that can detect extremist speech in spoken recordings. The system uses a multimodal classifier that combines speech-to-text transcription with analysis of emotional tone and vocal intonation to determine whether the content is hateful or extremist. Source code: `https://github.com/bewuwy/burtcore-junction`

2024 – 2025 — **NetNote** - a team project I created for my CSE course at the university. The program is a note taking application with various customizations (e.g., themes, tags, and note collections) and server syncing. It was made in Java using JavaFX and Spring.

2024 — **QuickQueue** - a demo of a queue system with numbered tickets written in Next.js and using PostgreSQL for database. Source code: `https://github.com/bewuwy/quick-queue`

2023 — **Battlecode 2023** - I worked together in a group to program an autonomous bot (in Java) for the tournament organised by MIT. Source code: `https://github.com/bewuwy/bc-23`

Neo Anki Leaderboard - a simple extension for the Anki program adding a leaderboard written in Python. I also created a website in SvelteKit using a PostgreSQL database to host the leaderboard online. Source code: `https://github.com/bewuwy/Neo-Anki-Leaderboard`

2022 – 2023 — **revers.io** - an online implementation of the Reversi board game supporting multiplayer games. The website was designed in Figma, written in Angular, and uses Google's Firebase database. Source code: `https://github.com/bewuwy/revers.io/`

2022 — **Airplane boarding simulator** - I created this simulator in Python using PyGame for a math modeling problem I had to solve. Source code: `https://github.com/bewuwy/airplane-boarding`

Academic experience

2025 – 2027 — **Honours Project** - "Reuse possibilities of mobile devices at the end of their lifespan: A look at the costs and benefits associated". For the honours part of my bachelor studies, I have started a research project focused on sustainability in computer science. The research aims to find whether it is reasonable to use old smartphones or laptops and their compute power to host real services and what is the best way to do so.

2023 – 2024 — **Mathematics Extended Essay** - "Exploring mathematics behind the Cody Dock bridge". I wrote a 30-page analysis of an unusual bridge located in London and how catenary curves were used in its construction. The topic of my research was calculating the road that the bridge "rolled on" using calculus and Python-based approximations. This work was part of my International Baccalaureate diploma. Link to the essay: `https://bewu.dev/ee-pdf`

2023 — **Warsaw Science Slam** - My talk "The quietest theft of the century - How AI stole art." was about the development of image generating models. I explored how the generative models like Stable Diffusion go from text to "art" and what are the moral implications associated with their development.